

Vibration Isolator using Hybrid Reluctance Actuator toward Quasi-zero Stiffness

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Abstract: This paper proposes a sample-tracking vibration isolator that integrates a flexure-guided hybrid reluctance actuator (HRA) with a negative stiffness for quasi-zero stiffness (QZS). In a sample-tracking vibration isolator, for example, for inline metrology, an actuator carries and positions an instrument with respect to a sample. By maintaining the distance between the instrument and the sample by feedback control, the vibrations between them are rejected for measurements in vibrational environments. To achieve high vibration isolation and rejection, the proposed sample-tracking vibration isolator uses the negative stiffness that is magnetically realized and cancels a flexure stiffness for QZS. Due to the rigid magnetic components of the HRA such as a permanent magnet and ferromagnetic cores, a high second mechanical frequency of 2.1 kHz is realized. Consequently, a high closed-loop control bandwidth of 806 Hz is achieved with feedback control. In order to validate the effectiveness of the rigid negative stiffness of the HRA, the negative stiffness is tuned to cancel a flexure stiffness at experiments. Experimental results show that transmissibility at a low frequency of 10 Hz is significantly decreased from -34 dB to -60 dB by a factor of 20 and successfully demonstrate high vibration isolation performance of the sample-tracking vibration isolator using the negative stiffness of the HRA.

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Keywords: Mechatronic Systems, Motion control, Vibration isolation, Quasi-zero stiffness

1. INTRODUCTION

Vibrations transmitted from the environment are problematic for high-precision manipulation and instrumentation (e.g. atomic force microscopy (AFM)). These vibrations can be compensated by using vibration isolators, and their requirements depend on applications. A typically required bandwidth of a vibration isolator to compensate for floor vibrations is from 1 Hz to 80 Hz in a building [Amick et al. (2005)]. In the case of inline metrology in Fig. 1 [Wertjan et al. (2022)] [Csencsics et al. (2022)], vibration isolation at the end effector of a robot is required to bring and operate instruments in a production line, and more challenging requirements are imposed to compensate for both floor vibrations and disturbances generated by the robot itself. This application requires a compact vibration isolator with a larger isolation bandwidth of at least 1 Hz - 200 Hz and better isolation performance [Csencsics et al. (2022)], for which suitable technologies are necessary.

In the case of typical passive vibration isolators, a tabletop is suspended by springs with a positive stiffness (e.g. air springs). Due to the dynamics that decouple the tabletop from the floor beyond the natural frequency, the tabletop needs to be relatively heavy and bulky to isolate vibrations at low frequencies [Yan et al. (2020)] [Yuan et al. (2020)]. Vibration isolators can be light and compact by using active vibration isolation. However, it has difficulty to isolate vibrations at low frequencies, where the signal-

to-noise ratio of vibration sensors (e.g. geophones) is degraded [Brincker et al. (2005)] [Beijen et al. (2018)]. Consequently, conventional passive or active vibration isolators are not suitable for vibration isolation at a robot end effector for robot-based inline metrology.

Sample-tracking vibration isolators [Ito et al. (2019a)] have a potential to solve the above problems. Fig. 1 shows a robot that carries an AFM with sample-tracking vibration isolation along the vertical axis. A displacement sensor on the mover measures the distance to a sample, and it is actively maintained by an actuator with a feedback controller. Consequently, vibrations between the mover and the sample are rejected, and an AFM on the mover can measure a sample surface precisely in vibrational environments.

As the actuator of a sample-tracking vibration isolator, Lorentz actuators (e.g. voice coil actuators) are often used [Wertjan et al. (2022)] for their high linearity [Ito and Schitter (2016)] and zero-stiffness property to isolate vibrations. However, their motor constant (force-to-current ratio) is relatively low. As a result, a large coil current is required to generate a high actuation force or to compensate for gravity, wasting energy in the form of heat. This problem can be solved by using a hybrid reluctance actuator (HRA) [Ito et al. (2019a)], the motor constant of which can be a few times higher than that of comparable Lorentz actuators for energy efficiency [Ito et al. (2019b)]. However, HRAs themselves are unstable and requires me-

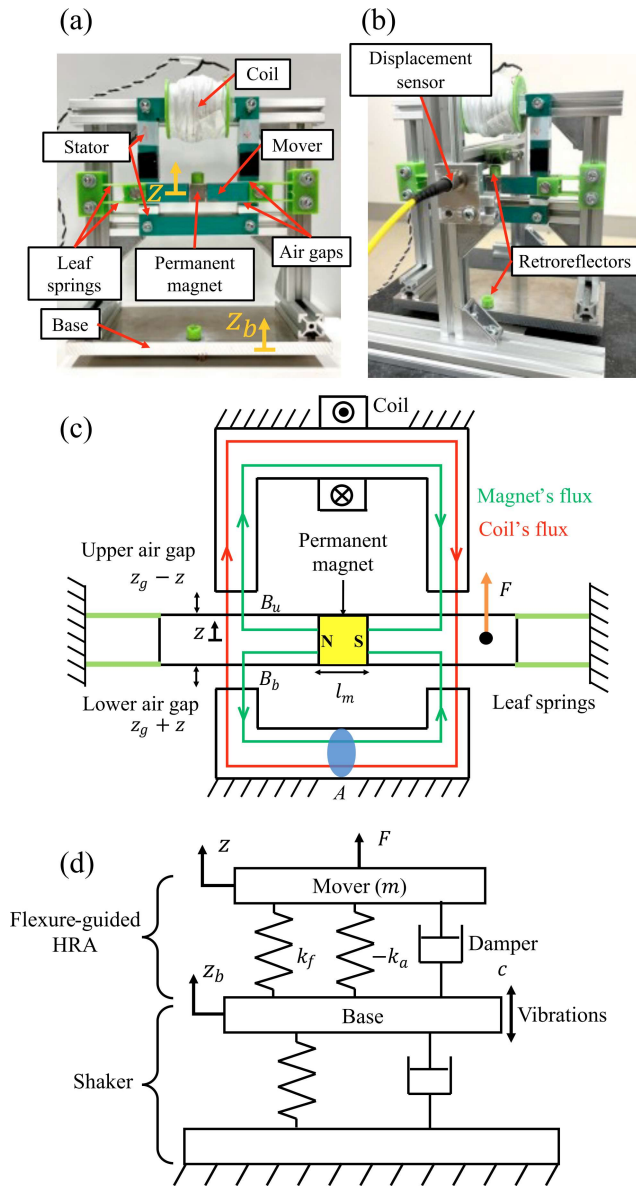


Fig. 2. Sample-tracking vibration isolator using a flexure-guided HRA. (a) Photograph. (b) Photograph with a displacement sensor. (c) Illustration showing the actuation principle. (d) Lumped mass model of the sample-tracking tracking vibration isolator on the shaker.

leakage and fringing flux, respectively. Dependent on the mover position z , the magnetic flux ϕ_m is divided into two fluxes through the upper air gaps ϕ_{mu} and lower air gaps ϕ_{ml} , as follows:

$$\phi_{mu} = \frac{\phi_m R_l}{R_u + R_l}, \quad \phi_{ml} = \frac{\phi_m R_u}{R_u + R_l}. \quad (3)$$

Since the magnetic reluctance of the permanent magnet is sufficiently higher than that of the air gaps in (1) for $l_m = 15\text{ mm}$, the permanent magnetic flux generated by the coil ϕ_c does not go through the magnet and is given by

$$\phi_c = \frac{NI}{2(R_u + R_l)}, \quad (4)$$

with the winding number $N = 250$ and the coil current I . By applying the Maxwell stress tensor [Furlani (2001)], the actuation force of the HRA is obtained as

$$F = \frac{(\phi_{mu} + \phi_c)^2 - (\phi_{ml} - \phi_c)^2}{\mu_0 A} = K_m(z)I + k_a(z)z, \quad (5)$$

where K_m and k_a are the motor constant and the actuator stiffness, respectively, and they are given by

$$K_m(z) = \frac{\mu_0 AN \lambda H_c l_m}{2(l_m z_g + z_g^2 - z^2)}, \quad (6)$$

$$k_a(z) = \frac{\mu_0 A \lambda^2 H_c^2 l_m^2 z_g}{(l_m z_g + z_g^2 - z^2)^2}. \quad (7)$$

3.2 Lumped mass model and dynamics

To analyze the influence of the actuator stiffness $k_a(z)$ on vibration isolation, the sample-tracking vibration isolator is described by a lumped mass model in Fig. 2 (d) with a mover mass m of 190 g. The damping coefficient c is usually sufficiently smaller than the critical damping. The flexures are designed and manufactured by a 3D printer (Adventurer3, FLASHFORGE, Zhejiang, China) and its stiffness k_f is given by [Ito et al. (2016)]

$$k_f = \frac{nEwh^3}{L^3}, \quad (8)$$

with Young's modulus $E = 3.15\text{ GPa}$. The number of flexures $n = 4$, the flexure width $w = 19\text{ mm}$, the flexure height $h = 1.2\text{ mm}$, and the flexure length $L = 16.8\text{ mm}$ are adjusted such that k_f and k_a are balanced when they are implemented, resulting in $k_f = 87.2\text{ N/mm}$. An equation of motion about the mover is obtained as follows

$$m\ddot{z} + c(\dot{z} - \dot{z}_b) + k_f(z - z_b) = F. \quad (9)$$

The Laplace transforms of (5) and (9) gives

$$z(s) = P_a(s)I(s) + P_t(s)z_b(s), \quad (10)$$

where $P_a(s)$ and $P_t(s)$ are transfer functions from I to z and from z_b to z , respectively, as shown in Fig. 3. They are represented by

$$P_a = \frac{z(s)}{I(s)} = \frac{K_m(z)}{ms^2 + cs + k_f - k_a(z)}, \quad (11)$$

$$P_t = \frac{z(s)}{z_b(s)} = \frac{cs + k_f - k_a(z)}{ms^2 + cs + k_f - k_a(z)}. \quad (12)$$

The transfer function $P_t(s)$ represents the transmissibility of the sample-tracking vibration isolator without feedback control. Note that an unstable pole occurs with $k_f < k_a$. Thus, feedback control is designed for stability to fine-tune k_a . In addition, the first resonant frequency ω_n about the mover is given from (12) as follows:

$$\omega_n = \sqrt{\frac{k_f - k_a(z)}{m}}. \quad (13)$$

This equation indicates that ω_n is decreased when k_a cancels k_f . The decrease of the first resonant frequency is used to validate the effectiveness of the negative stiffness for QZS in Section 5.

For stability and vibration rejection, feedback control is designed in Section 4. The transmissibility with a feedback

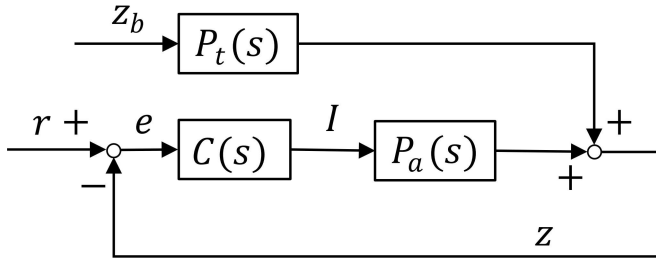


Fig. 3. Block diagram of the sample-tracking vibration isolator.

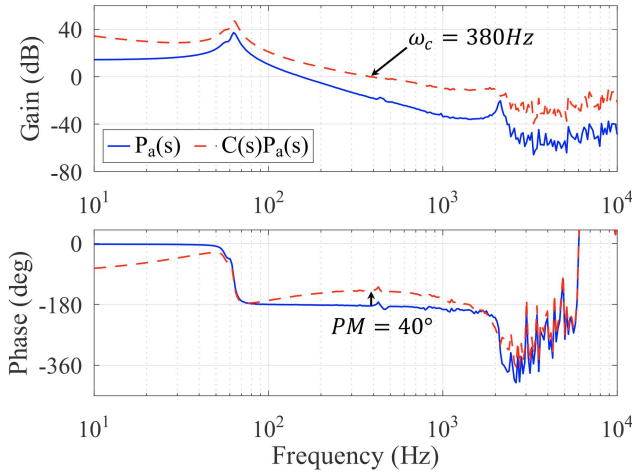


Fig. 4. Measured Bode plot of the plant $P_a(s)$ from the coil current I to the mover position z and an open-loop transfer function $C(s)P_a(s)$.

controller $C(s)$ is derived by substituting $(r - z(s))C(s)$ for I : [Ito and Schitter (2016)]

$$\frac{z(s)}{z_b(s)} = \frac{P_t(s)}{1 + P_a(s)C(s)} = S(s)P_t(s), \quad (14)$$

where $S(s)$ is a sensitivity function of the feedback loop, which is equivalent to the transfer function from a position reference r to a position error e . For high vibration isolation performance, typically $S(s)$ has high-pass characteristics with a low gain at low frequencies.

Note that the damper c can transmit vibrations z_b particularly at high frequencies even in the case of $k_a = k_f$ (see Fig. 2 (d)). The influence of c will be investigated as a part of future work.

4. CONTROL DESIGN

For stability and vibration rejection, this section presents the design of the feedback controller $C(s)$.

4.1 Frequency response

To design the controller $C(s)$, a Bode plot from the coil current I to the mover position z in Fig. 3 is measured, as the results are shown in Fig. 4. The Bode plot shows low-pass characteristics with a first resonance at 63 Hz. A second resonance is at 2.1 kHz, which is successfully far higher than the first resonant frequency for a high control bandwidth, as intended.

4.2 Tamed PID controller with notch filter

For high vibration isolation performances, the feedback controller needs to realize a control bandwidth much higher than the first resonant frequency at 63 Hz. For that purpose, a tamed PID controller with a notch filter [Munnig Schmidt et al. (2020)] as $C(s)$ is used.

As the first step of control design, a notch filter with a notch frequency at 2.1 kHz is implemented, trimming the peak gain of the second resonance for a gain margin. The notch filter is represented by

$$C_{notch}(s) = \frac{s^2 + 2D\zeta\omega_{nf} + \omega_{nf}^2}{s^2 + 2\zeta\omega_{nf} + \omega_{nf}^2}, \quad (15)$$

where D , ζ , and ω_{nf} are parameters to determine the depth, the width, and the frequency of the notch, respectively, and they are tuned to the values shown in Table 2.

Table 2. Parameters of the notch filter in $C(s)$.

Parameter	Value
D	-22 dB
ζ	0.13
ω_{nf}	$2\pi \times 2.1 \times 10^3$ rad/s

For a high control bandwidth beyond the first resonant frequency, the tamed PID controller is configured [Munnig Schmidt et al. (2020)] as follows

$$C_{pid}(s) = g_{pid} \frac{(s + \frac{\omega_c}{10})(3s + \omega_c)}{s(s + 3\omega_c)}, \quad (16)$$

where g_{pid} and ω_c are a gain and a desired crossover frequency, respectively. They are tuned to maximize ω_c with a sufficient stability margin of more than 40° , and the tuned parameters are $\omega_c = 380$ Hz and $g_{pid} = 17.5$ dB. By combining the tamed PID controller and the notch filter, the resulting controller is

$$C(s) = C_{pid}(s)C_{notch}(s). \quad (17)$$

The designed controller is discretized by a pole-zero matching method [Franklin et al. (1998)] and implemented by the rapid prototyping control system. For evaluation of the closed-loop stability, the open-loop transfer function $C(s)P_a(s)$ in Fig. 3 is measured as shown in Fig. 4. It indicates the controller $C(s)$ achieves the open-loop crossover frequency of 380 Hz with a sufficient phase margin of 40° and a gain margin of 9.28 dB.

For validation of the control design, the complementary sensitivity function $T(s)$ from the position reference r to the mover position z is measured as shown in Fig. 5. It is visible that the achieved control bandwidth (-3 dB) is 806 Hz, due to the high second resonant frequency of 2.1 kHz.

5. EXPERIMENTAL RESULT

To investigate the influence of the actuator stiffness k_a , a Bode plots of the plant $P_a(s)$ are measured at several mover positions z . To set the mover position z , the feedback control is used and an offset signal is superimposed to a sine signal of the position reference r . The results

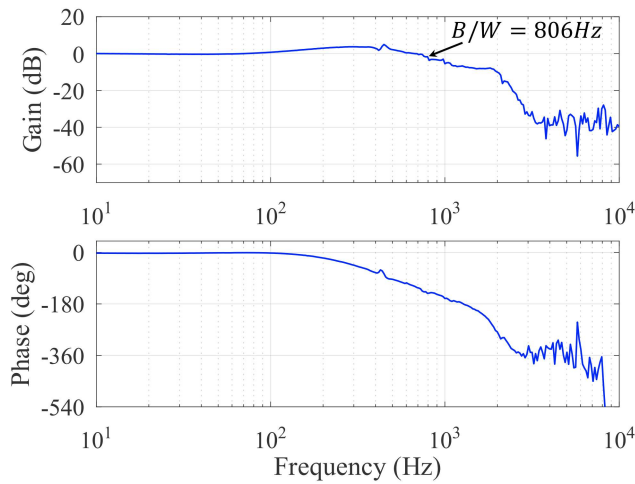


Fig. 5. Measured complementary sensitivity function $T(s)$ from the position reference r to the mover position z to validate the control design.

are shown in Fig. 6, it is visible that the first resonant frequency is decreased when the mover position is changed from $z = 0 \mu\text{m}$ to $z = -385 \mu\text{m}$. At $z = -430 \mu\text{m}$, however, the Bode plot shows low-pass characteristics without a resonance peak and a phase shift of 180° . Such a response occurs at $k_a > k_f$ [Ito et al. (2019b)]. This means that QZS can be realized at a point between $z = 0 \mu\text{m}$ and $-430 \mu\text{m}$. In fact, by setting z to $-385 \mu\text{m}$, the first resonant frequency is decreased from 65 Hz to less than 20 Hz for a -40 dB/dec mass line over a large frequency band up to the second resonant frequency of 2.1 kHz. In future work, fine tuning of z will be carried out by using a better network analyzer.

Overall, Fig. 6 clearly shows the effectiveness of the HRA's negative stiffness k_a to cancel the flexure stiffness k_f by lowering the first resonant frequency. Furthermore, the second resonant frequency around a high frequency of 2.1 kHz indicates the rigidity of the magnetically realized negative stiffness, as intended in the mechanical and magnetic design.

To demonstrate the effectiveness of the stiffness cancellation, the transmissibility with the feedback control z/z_b is measured by varying the mover position in the same manner as Fig. 6. The results are shown in Fig. 7. At $z = 0 \mu\text{m}$ with the first resonant frequency of 65 Hz, the transmissibility is -34 dB at 10 Hz and -23 dB at 200 Hz only. By tuning z , the transmissibility is well-improved. Particularly at $z = -385 \mu\text{m}$, it is approximately -60 dB at 10 Hz and -39 dB at 200 Hz. This high vibration isolation performance is realized by cancelling k_f by k_a to decrease the transmission of the vibration z_b .

In summary, the experiments successfully demonstrate that the magnetically-realized negative stiffness k_a is rigid for high second resonant frequency of 2.1 kHz to achieve a high control bandwidth of 806 Hz and enables to decrease the mover-stator stiffness for high vibration isolation performances.

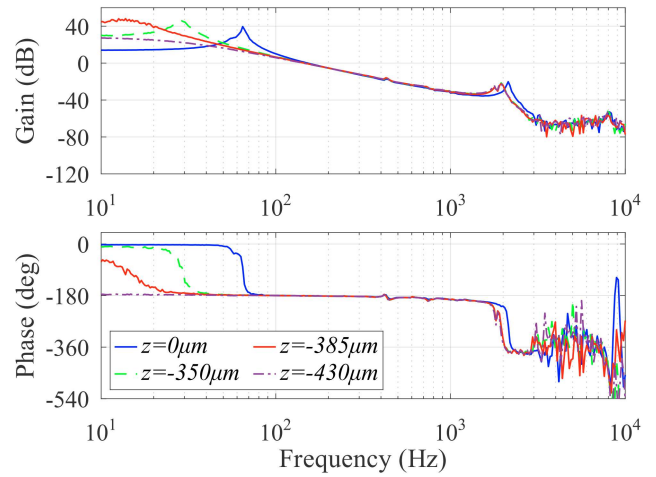


Fig. 6. Measured Bode plot of the plant $P_a(s)$ at several mover positions.

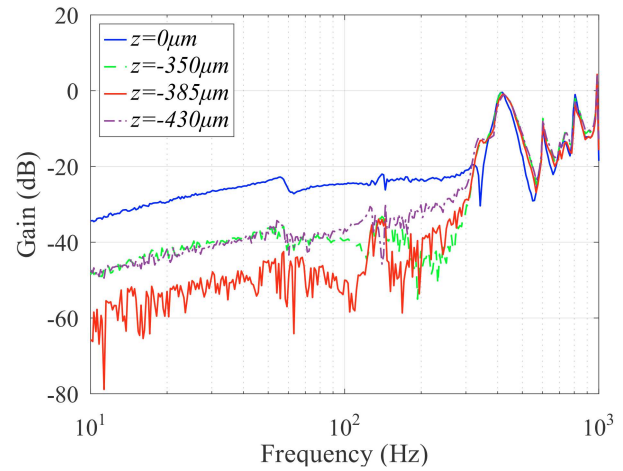


Fig. 7. Measured transmissibility with the feedback control of the proposed sample-tracking vibration isolator from z_b to z at several mover positions.

6. CONCLUSION

For vibration isolation at the tip of a robot for inline metrology, this paper proposes a sample-tracking vibration isolator with a flexure-guided HRA. To simultaneously decrease the transmission of the vibrations from the stators and achieve a high control bandwidth for vibration rejection, the HRA magnetically realizes a negative stiffness to cancel the flexure stiffness for QZS. Due to the rigid magnetic components such as a permanent magnet and ferromagnetic cores, the second mechanical resonant frequency is successfully set to a high frequency of 2.1 kHz. Consequently, a high control bandwidth of 806 Hz is achieved for vibration rejection by a tamed PID controller with a notch filter that compensates for the second resonance. At an experiment, the negative stiffness is tuned to match with the flexure stiffness, and the transmissibility is measured for validation. By tuning the negative stiffness, the transmissibility at 10 Hz is significantly decreased from -34 dB to -60 dB by a factor of 20. The experimental results successfully demonstrate the effectiveness of the rigid negative stiffness for high control bandwidth and

high vibration isolation performance. Future work includes development of algorithms to automatically fine-tune the negative stiffness.

REFERENCES

- Amick, H., Gendreau, M., Busch, T., and Gordon, C. (2005). Evolving criteria for research facilities: vibration. volume 5933, 593303. International Society for Optics and Photonics, SPIE.
- Beijen, M., Heertjes, M., Van Dijk, J., and Hakvoort, W. (2018). Self-tuning mimo disturbance feedforward control for active hard-mounted vibration isolators. *Control Engineering Practice*, 72, 90–103.
- Brincker, R., Lagö, T., Andersen, P., and Ventura, C. (2005). Improving the classical geophone sensor element by digital correction. In *Conference Proceedings*. Society for Experimental Mechanics.
- Csencsics, E., Thier, M., Ito, S., and Schitter, G. (2022). Supplemental peak filters for advanced disturbance rejection on a high precision endeffector for robot-based inline metrology. *IEEE/ASME Transactions on Mechatronics*, 27(4), 2258–2266.
- Franklin, G.F., Powell, J.D., and Workman, M.L. (1998). *Digital Control of Dynamic Systems (3rd Edition)*. Addison-Wesley Reading, MA.
- Furlani, E.P. (2001). Permanent Magnet and Electromechanical Devices. *Academic Press*.
- Ito, S., Cigarini, F., Unger, S., and Schitter, G. (2016). Flexure design for precision positioning using low-stiffness actuators. *IFAC-PapersOnLine*, 49(21), 200–205.
- Ito, S., Lindner, B., and Schitter, G. (2019a). Sample-tracking vibration isolation with hybrid reluctance actuators for inline metrology. *IFAC-PapersOnLine*, 52(15), 537–542.
- Ito, S. and Schitter, G. (2016). Comparison and classification of high-precision actuators based on stiffness influencing vibration isolation. *IEEE/ASME Transactions on Mechatronics*, 21(2), 1169–1178.
- Ito, S., Troppmair, S., Lindner, B., Cigarini, F., and Schitter, G. (2019b). Long-range fast nanopositioner using nonlinearities of hybrid reluctance actuator for energy efficiency. *IEEE Transactions on Industrial Electronics*, 66(4), 3051–3059.
- Kluk, D.J., Boulet, M.T., and Trumper, D.L. (2012). A high-bandwidth, high-precision, two-axis steering mirror with moving iron actuator. *Mechatronics*, 22(3), 257–270. Special Issue on Mechatronic Systems for Micro- and Nanoscale Applications.
- Munnig Schmidt, R., Schitter, G., Rankers, A., and van Eijk, J. (2020). The Design of High Performance Mechatronics. *Delft University Press, 3rd revised edition*.
- Wang, P. and Xu, Q. (2016). Design of a flexure-based xy precision positioning stage with constant force output. In *In Annual Conference of the IEEE Industrial Electronics Society*, 524–529.
- Wertjan, D., Csencsics, E., Kern, T., and Schitter, G. (2022). Bringing the lab to the fab: Robot-based inline measurement system for precise 3-d surface inspection in vibrational environments. *IEEE Transactions on Industrial Electronics*, 69(10), 10666–10673.
- Yan, G., Zou, H.X., Wang, S., Zhao, L.C., Gao, Q.H., Tan, T., and Zhang, W.M. (2020). Large stroke quasi-zero stiffness vibration isolator using three-link mechanism. *Journal of Sound and Vibration*, 478, 115344.
- Yuan, S., Sun, Y., Zhao, J., Meng, K., Wang, M., Pu, H., Peng, Y., Luo, J., and Xie, S. (2020). A tunable quasi-zero stiffness isolator based on a linear electromagnetic spring. *Journal of Sound and Vibration*, 482, 115449.